DOMAIN 5 STUDY MATERIAL QUESTIONS

Question 1

Question: Which of the following statements is the correct definition of the process of adsorption?

- a. Adsorption is the process of removing a component of a liquid mixture into a gas stream.
- b. Adsorption is the process of removing a component of a liquid mixture into another liquid.
- c. Adsorption is the process of removing a pure gas or one or more components of a gaseous mixture into a liquid stream.
- d. Adsorption is the process of removing a pure gas or one or more components of a gaseous mixture into a solid bed.

Correct Answer: Select (d)
Explanation: Air purifying respirators remove the contaminants in the air by adsorption. Activated charcoal is a good example of an adsorbent.

Question 2

Question: Chemical hazardous wastes must be segregated from incompatibles. Chemical wastes should be segregated by general groupings. Which of the following is NOT part of chemical groupings for segregation of hazardous waste?

- a. Mineral acids
- b. Alkenes
- c. Oxidizers
- d. Caustics

Correct Answer: Select (b)
Explanation: The other groupings are: flammables, non-halogenated organic solvents, halogenated organic solvents, organic acids, carcinogens, mercury and mercury compounds, and general "dry" lab chemicals.

Question 3

Question: Hazardous wastes under RCRA are divided into two major categories: characteristic wastes and listed wastes. Which of the following is NOT the property of a characteristic waste?

- a. Ignitability
- b. Permeability
- c. Reactivity
- d. Toxicity

Correct Answer: Select (b)
Explanation: The fourth trait for characteristic hazardous waste is corrosivity. Toxicity is determined by Toxicity Characteristic Leaching Procedure (TCLP). Under RCRA, ignitable wastes are defined by their flash points.

Question 4

Question: Air pollutants are classified into "primary" and "secondary pollutants." Which of the following is an example of a secondary air pollutant?

- a. Carbon monoxide in the air
- b. Sulfur oxides in the air
- c. Nitric acid in the air
- d. Nitrogen oxides in the air

Correct Answer: Select (c)
Explanation: Secondary pollutants are formed as a result of reaction between a primary pollutant and moisture and other elements in the atmosphere. For example, nitric acid is formed as a result of reaction between nitrogen oxides (primary pollutants) and moisture in the atmosphere. Also sulfuric acid which is formed as a result of reaction between sulfur oxides and moisture is considered a secondary air pollutant.

Question 5

Question: Which of the following is NOT a means of control for air pollutants?

- a. Combustion
- b. Distillation
- c. Absorption
- d. Adsorption

Correct Answer: Select (b)
Explanation: Combustion is used when the air pollutants are organic gases and vapors. Combustion converts the pollutants to carbon dioxide and water.

Absorption: The harmful gases are removed by a suitable liquid in scrubbers.

Adsorption: The harmful gases are removed on the surface of a suitable porous solid.

Question 6

Question: All of the following are examples of proper segregation of hazardous wastes EXCEPT:

- a. Segregate oxidizers from organics and flammables
- b. Segregate cyanides from acids
- c. Segregate aldehydes from ketones
- d. Segregate acids from bases

Correct Answer: Select (c)
Explanation: Oxidizers can release oxygen that can make an ongoing fire in flammables much worse. Acid-base reactions are heat generating and the reaction between acids and cyanides can release highly toxic gases.

Question 7

Question: Which of the following steps help reduce storm water pollution from hazardous wastes?

- a. Hazardous wastes must be stored in a sealed container which is made of suitable materials of construction and must be properly labeled.
- b. Storage, transportation, and disposal must comply with provisions of federal, state, and local laws and regulations.
- c. Waste containers must be stored in containment facilities
- d. All of the above

Correct Answer: Select (d)
Explanation: The steps outlined in (a), (b), and (c) are designed to minimize storm water pollution from hazardous wastes.

Question 8

Question: Ground-level or "bad" ozone is not emitted directly into the air, but is created by chemical reactions. Which of the following chemicals react to produce "bad" ozone?

- a. Reaction between oxides of nitrogen (NOx) and volatile organic compounds in the presence of sunlight.
- b. Reaction between oxygen molecule (O2) and oxygen atom (O) in presence of ultra violet light.
- c. Reaction between oxygen and volatile organic compounds catalyzed by lighting
- d. Reaction between sulfur oxide (SOx) and hydrocarbon gases in presence of sunlight

Correct Answer: Select (a)
Explanation: Ground-level or "bad" ozone can be generated by the chemical reaction between oxides of nitrogen and volatile organic compounds. The reaction is catalyzed by sunlight.

Question 9

Question: Which of the following processes is NOT a source of lead emissions into the air?

- a. Ore and metal processing operations
- b. Waste incinerations
- c. Scrubbing operations
- d. Battery manufacturing operations

Correct Answer: Select (c)
Explanation: Other sources of lead emissions into the air include utilities plants and piston-engine air craft operating on leaded aviation fuel.

Question 10

Question: The EPA has identified six air pollutants as "criteria" air pollutants. Which of the following is NOT a member of "criteria" air pollutants as identified by the EPA?

- a. Carbon monoxide
- b. Nitrogen oxides
- c. Ground-level ozone
- d. Carbon dioxide

Correct Answer: Select (d)
Explanation: The other three air pollutants identified as "criteria" pollutants are sulfur oxides, lead, and particulate matter.

Question 11

Question: The EPA has identified six air pollutants because it regulates them by developing human health-based and/or environmentally-based criteria for setting permissible levels. These six air pollutants are known as:

- a. NESHAP pollutants
- b. Criteria Pollutants
- c. Highly toxic Pollutants
- d. All of the above

Correct Answer: Select (b)
Explanation: The six air pollutants are identified by the EPA as "criteria" pollutants. These are carbon monoxide, nitrogen oxides, ground-level ozone, sulfur oxides, lead, and particulate matter.

Question 12

Question: Unconfined aquifers are also called:

- a. Water table aquifers.
- b. Saturated aquifers.
- c. Confining bed aquifers.
- d. Unsaturated aquifers.

Correct Answer: Select (a)
Explanation: Unconfined aquifers are usually in unconsolidated material and the water table depth varies seasonally. The water table tends to rise in the winter when there is more rainfall, and fall in the summer due to less rainfall infiltration.

Question 13

Question: The instrument used to measure water table elevation is called a:

- a. Hydrometer.
- b. Piezometer.
- c. Aquimeter.
- d. Darcymeter.

Correct Answer: Select (b)
Explanation: A piezometer is used to measure water table elevation.

Question 14

Question: The ratio of volume of voids to the total volume of rock or soil is called:

- a. Void density.
- b. Homogeneity.
- c. Void specific gravity.
- d. Total porosity.

Correct Answer: Select (d)
Explanation: This ratio is called the total porosity of rock or soil. It is expressed as a percentage or fraction. Porosity is a measure of the maximum amount of water a rock or soil can hold when it is saturated with water. For example, gravel porosity has a range of 25 to 40 percent.

Question 15

Question: One of the important characteristics of aquifers is "specific yield". Which of the following statements BEST describes this term?

- a. It is a measure of how quickly water will pass through the aquifer.
- b. It is the volume of water that is retained on soil or rock after gravity drainage.
- c. It is the volume of water that will drain by gravity from a specific volume of soil or rock.
- d. All of the above.

Correct Answer: Select (c)
Explanation: Statement (b) is a description of specific retention. Statement (a) is a description of hydraulic conductivity. The porosity of a rock or soil is the sum of specific yield and specific retention.

Question 16

Question: Which of the following is used to calculate the velocity and travel times of pollutants in groundwater?

- a. Raoult's Law
- b. Fick's Law
- c. Newton's second law of motion
- d. Darcy's Law

Correct Answer: Select (d)
Explanation: Darcy's Law is useful in quantifying the movement of pollutants through underground waters. It relates the discharge (volume per unit time) to the hydraulic conductivity of the aquifer, the area the water moves through horizontally, and a dimensionless friction or head pressure loss.

[Q =

rac{K \cdot A \cdot (H_1 - H_2)}{L}]

Where:

- (Q) = discharge, volume per unit time
- (K) = hydraulic conductivity
- (A) = cross sectional area
- (

rac{H_1 - H_2}{L}) = a dimensionless head pressure loss term

Question 17

Question: Which of the following is formed when ground water is pumped from an unconfined aquifer?

- a. Sinkholes
- b. Cone of depression
- c. Drawdown
- d. Confining bed

Correct Answer: Select (b)
Explanation: When water is pumped from a confined aquifer, a cone of depression is formed.

Question 18

Question: Which of the following is considered to be the primary source of ground water?

- a. Rainfall
- b. Snow
- c. Rivers
- d. Lakes

Correct Answer: Select (a)
Explanation: Rainfall is considered to be the primary source of groundwater.

Question 19

Question: Rainfall has a pH of about 5.7. This means that:

- a. Rainfall is slightly acidic.
- b. Rainfall can dissolve carbonates in the soil or rock.
- c. The ground water has a PH greater than the rainfall.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Rainfall generally has a pH less than 7.0 due to dissolved gases from the atmosphere. This makes rainfall slightly acidic. Therefore, rainfall can easily dissolve and react with carbonates and other minerals that are bases. This process increases the pH of the rainfall as it migrates and joins the underground waters.

Question 20

Question: An underground storage tank containing diesel fuel has been leaking and might have caused groundwater contamination. The monitoring well for sampling must be designed in such a way to provide means for sampling from the _____ of the aquifer.

- a. Bottom
- b. Mid-point
- c. Top
- d. Top, mid-point, and bottom (mix to make a representative sample)

Correct Answer: Select (c)
Explanation: Diesel fuel has a specific gravity of less than 1.0 and is insoluble in water. It will float on top of water. The sample should be taken from the top of the aquifer.

Question 21

Question: The unsaturated zone is an area directly below the soil surface where the soil particles are surrounded by both water and air. The unsaturated zone is also called the:

- a. Potentiometric zone.
- b. Confining bed.
- c. Vadose zone.
- d. Lithification zone.

Correct Answer: Select (c)
Explanation: The unsaturated zone is also called the Vadose zone. The saturated zone is the area beneath the unsaturated zone where the soil particles are completely surrounded by water.

Question 22

Question: A measure of the ability of an aquifer to transmit water is:

- a. Gravel.
- b. Specific retention.
- c. Specific velocity.
- d. Hydraulic conductivity.

Correct Answer: Select (d)
Explanation: Hydraulic conductivity is a measure of how fast water passes through an aquifer. The fluid properties and the properties of the aquifer materials are factored into the value of hydraulic conductivity.

Question 23

Question: Which of the following statements is true in regard to soil permeability?

- a. It is a measure of how easily water can move through an aquifer.
- b. The permeability value depends on the properties of the aquifer materials only.
- c. Permeability is a value usually obtained in a laboratory.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Statements in (a), (b), (c) are true in regard to permeability. Permeability is very similar to hydraulic conductivity. Usually, permeability is a value obtained in a laboratory, while hydraulic conductivity is measured in field.

Question 24

Question: The boundary between the saturated and unsaturated zones of soil is called the:

- a. Water table aquifer.
- b. Potentiometric surface.
- c. Lithified zone.
- d. Vadose zone.

Correct Answer: Select (b)
Explanation: A potentiometric surface is an area where the hydrostatic pressure exerted by the water table is equal to atmospheric pressure.

Question 25

Question: The process that reduces the solubility of hazardous constituents and reduces the exposed area that may allow the migration of the contaminants is called:

- a. Screening.
- b. Sedimentation.
- c. Stabilization.
- d. Flotation.

Correct Answer: Select (c)
Explanation: Most stabilization processes incorporate the waste into a solid matrix using a binding agent or polymers.

Question 26

Question: LDR is an acronym for which of the following?

- a. Laboratory Disposal Report
- b. Land Disposal Restrictions
- c. Lowest Dose Reported
- d. Lethal Dose Restrictions

Correct Answer: Select (b)
Explanation: LDR is an acronym for Land Disposal Restrictions. LDR restricted wastes are those that fall under the land disposal restriction standard under RCRA.

Question 27

Question: Contaminated wastewater and groundwater can be treated biologically to remove hazardous substances. The microorganisms for biological treatment may be classified as heterotrophic and autotrophic. Which of the following statements is true in regard to these classes of microorganisms?

- a. Heterotrophs use inorganic matter.
- b. Heterotrophs use organic matter.
- c. Autotrophs use organic matter.
- d. Both heterotrophs and autotrophs use organic matter.

Correct Answer: Select (b)
Explanation: Heterotrophs use organic matter and autotrophs use inorganic matter as the source of their nutrients in contaminated wastewater, groundwater, and soil.

Question 28

Question: Microorganisms may be categorized as aerobic and anaerobic according to their utilization of oxygen. Which of the following statements is true in regard to these classes of microbes?

- a. Anaerobic microbes use oxygen that is combined with other elements.
- b. Anaerobic microbes use free oxygen.
- c. Aerobic microbes use oxygen that is combined with other elements.
- d. Anaerobic microbes can use both free and combined oxygen.

Correct Answer: Select (b)
Explanation: Anaerobic microbes use oxygen that is combined with other elements. Examples are nitrates, carbonates, or sulfates.

Question 29

Question: Undesirable heavy metals are usually removed from a liquid waste by precipitation. The solubility of heavy metals in a liquid waste can be reduced by varying the ____.

- a. PH.
- b. Specific gravity.
- c. Conductivity.
- d. Ion exchange.

Correct Answer: Select (a)
Explanation: The pH is an important factor in removing heavy metals from a liquid waste stream. For example, the neutralization of an acidic waste can also cause the precipitation of heavy metals. The hydroxides of heavy metals are usually insoluble, so lime is commonly used for precipitating them.

Question 30

Question: Air stripping is commonly used to remove low concentration of hazardous materials from wastewater. Which of the following statements is true in regard to air stripping?

- a. Stripping towers may be induced-draft towers.
- b. In stripping towers, the air flows upward and the liquid waste flows downward.
- c. The gas-liquid system eventually reaches equilibrium based on Henry's Law.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are all true in regard to air stripping towers. The air-waste stream usually undergoes secondary treatment such as incineration. In steam stripping, live steam is injected into the liquid waste.

Question 31

Question: The removal of low concentrations of hazardous substances from aqueous waste using activated carbon is accomplished by which of the following mechanisms?

- a. Absorption
- b. Ion exchange
- c. Adsorption
- d. Floatation

Correct Answer: Select (c)
Explanation: Adsorption is the process of removing a component of a liquid or gas on a solid. Absorption is the process of removing a component into a liquid. Ion exchange is a reversible exchange of ions between liquid and solid phases.

Question 32

Question: Which of the following statements is NOT true in regard to dust particles?

- a. They are formed when particles migrate and condense from the surface of a molten solid into the air.
- b. They consist of mechanically-produced solid particles whose size can range from microscopic to visible.
- c. Dust particles can be inorganic or organic.
- d. The proper evaluation of dust hazards requires information regarding particle size, chemical composition, concentration, and factors that affect dispersion.

Correct Answer: Select (a)
Explanation: Statements (b), (c), and (d) are true in regard to dust. Statement (a) is a characteristic of fumes.

Question 33

Question: Cyclones are extensively used as an air pollution control device. Which of the following statements is NOT true in regard to cyclones?

- a. Cyclones operate to collect relatively large-size particulate matter from a gaseous stream through the use of centrifugal forces.
- b. Dust-laden gas is made to rotate in a decreasing diameter pathway, forcing solids to the outer edge of the gas stream for deposition into the bottom of the cyclone.
- c. Efficiencies of 90% in particle sizes of 10 microns or greater are possible.
- d. One of the disadvantages of cyclones is that operation at elevated temperatures is not possible.

Correct Answer: Select (d)
Explanation: Cyclones can operate at elevated temperatures. This is considered to be one of the advantages of cyclones.

Question 34

Question: Absorption and wet-scrubbing equipment are used extensively for air pollution control. All of the following statements are true in regard to this type of equipment EXCEPT:

- a. The goal is the removal of gases and particulate matter by causing the gaseous contamination to dissolve into the liquid stream and the solids to become entrained in the liquid.
- b. The rate of gas transfer into the liquid is dependent upon the solubility, mass transfer mechanism, and equilibrium concentration of the gas in solution.
- c. One of the advantages of this type of equipment is that high pressure drops are required.
- d. The rate of particulate matter collection at constant pressure drops is inversely proportional to the aerodynamic mean diameter of the particulate matter and scrubber droplets.

Correct Answer: Select (c)
Explanation: High pressure drops in absorption and wet-scrubbing equipment is a disadvantage.

Question 35

Question: Baghouses are used extensively for air pollution control. Which of the following statements is true in regard to this type of equipment?

- a. Baghouses use sieving, impaction, agglomeration, and electrostatic filtration principles to remove solids from a gaseous exhaust stream.
- b. Baghouses maximize the filtration area by configuring the fabric filter media into a series of long, small-diameter fabric tubes, referred to as bags.
- c. They are tightly packed into a housing wherein the dust-laden air moves across the bag fabric, thereby removing it from the gas stream and building up a filter cake which further enhances air cleaning.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are true in regard to baghouses.

Question 36

Question: Electrostatic precipitators are used extensively as air pollution control devices. Which of the following statements is true in regard to electrostatic precipitators?

- a. They use gaseous ions to charge particles, which are then moved through an electric field to be deposited onto charged collection plates.
- b. Collected particulate material is removed by rapping or washing the plates.
- c. To produce the free ions and electric field, high internal voltages are required.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are true in regard to electrostatic precipitators. Since the collection process does not rely on mechanical processes such as sieving or impaction, but rather electrostatic forces, the internal gas passages within a precipitator are relatively open with small pressure drops and lower energy costs to move the gas stream. Electrostatic precipitators are best suited for removing small particles (less than 1 micron) from the air stream. Electrostatic precipitators have very high removal efficiency (99%) for particulate matters.

Question 37

Question: An environmental Life-Cycle Assessment (LCA):

- a. Is a tool that can be used to evaluate the potential environmental impacts of a product, material, process, or activity
- b. Helps to promote the responsible design and redesign of products and processes
- c. Allows businesses to make product improvements through environmentally sound process, material, and design choices
- d. All of the above

Correct Answer: Select (d)
Explanation: LCA also gives businesses governmental incentives with the goal of obtaining environmental benefits.

Question 38

Question: Vapor-cloud dispersion models used for predictions of migration of hazardous materials into the atmosphere require input of data. All of the following are examples of data input into the dispersion models EXCEPT:

- a. Meteorological conditions
- b. Terrain evaluation at the source location
- c. Ambient air temperature
- d. Absorption coefficient

Correct Answer: Select (d)
Explanation: Other data that may be required include but are not limited to atmospheric turbulence, cloud cover, and solar radiation. The two types of vapor-cloud dispersion models are plume and puff.

Question 39

Question: At the simplest level, treatment of sewage and most wastewaters is carried out through which of the following?

- a. Distillation
- b. Absorption
- c. Crystallization
- d. Separation of solids from liquids

Correct Answer: Select (d)
Explanation: At the simplest level, treatment of sewage and most wastewaters is carried out through separation of solids from liquids, usually by sedimentation. By progressively converting dissolved material into solids, usually a biological floc, which is then settled out, an effluent stream of increasing purity is produced.

Question 40

Question: Wastewater comes in three main types. Which of the following does not belong to the three main types of wastewater?

- a. Blackwater
- b. Graywater
- c. Bluewater
- d. Yellow water

Correct Answer: Select (c)
Explanation: Wastewater comes in three main types namely Blackwater, Graywater and Yellow water. This is wastewater that originates from toilet fixtures, dishwashers, and food preparation sinks. It is made up of all the things that you can imagine going down the toilets, bath and sink drains.

Question 41

Question: REACH is an acronym for Registration, Evaluation, Authorization and Restriction of Chemicals. Which of the following statements is correct regarding REACH?

- a. It is enforced by OSHA under Hazards Communication standard (29CFR1910.1200).
- b. It is a European Union regulation.
- c. It is an EPA regulation enforced under Toxic Substances Control Act (TSCA).
- d. It is part of OSHA's voluntary standards.

Correct Answer: Select (b)
Explanation: Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) is a European Union regulation dating from 18 December 2006. REACH addresses the production and use of chemical substances and their potential impacts on both human health and the environment.

Question 42

Question: Which of the following statements is correct regarding the Restriction of Hazardous Substances (RoHS)?

- a. It is a legislation controlling hazardous substances throughout their whole life cycle.
- b. It is a legislation which focuses on hazardous substances in electrical and electronic equipment.
- c. It is an EPA regulation enforced under Land Disposal Restrictions (LDR).
- d. It is part of the Globally Harmonized System (GHS) for hazardous substances.

Correct Answer: Select (b)
Explanation: REACH is a horizontal legislation controlling the risks associated with chemical substances throughout their whole life cycle, while RoHS is a product-specific vertical legislation focuses on hazardous substances in electrical and electronic equipment (EEE).